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(54) **DEMISTER, ABSORPTION LIQUID ABSORBING TOWER, AND DEMISTER PRODUCTION METHOD**

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B01D 53/18 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC ***B01D 53/1475*** (2013.01); ***B01D 53/1418*** (2013.01); ***B01D 53/18*** (2013.01); ***B01D 53/62*** (2013.01); ***B01D 53/78*** (2013.01)

(58) **Field of Classification Search**

CPC B01D 47/06; B01D 47/14; B01D 53/14; B01D 53/1475; B01D 53/18; B01D 53/185

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(57) **ABSTRACT**

Provided are a demister, an absorption liquid absorbing tower, and a demister production method that enable efficient collection of mist. A demister for collecting a mist containing CO_2 absorption liquid, the demister comprising a plurality of laminates each including a first layer in which a plurality of linear structures having the axial direction aligned with a first direction are arranged in parallel to a second direction orthogonal to the first direction and a second layer in which a plurality of linear structures having the axial direction aligned with a direction different from the

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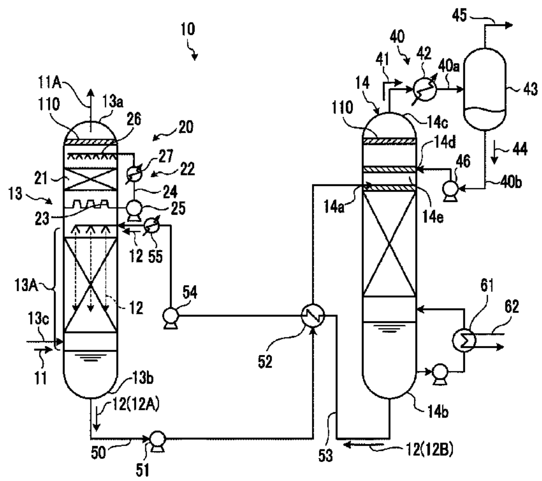

first direction are arranged in parallel in a direction orthogonal to said axial direction, wherein the laminates are stacked in a direction orthogonal to both the first and second directions.

5 Claims, 7 Drawing Sheets

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B01D 53/62 (2006.01)
B01D 53/78 (2006.01)

(58) **Field of Classification Search**
USPC 55/440, 442, 444, 450, 464, 465
See application file for complete search history.

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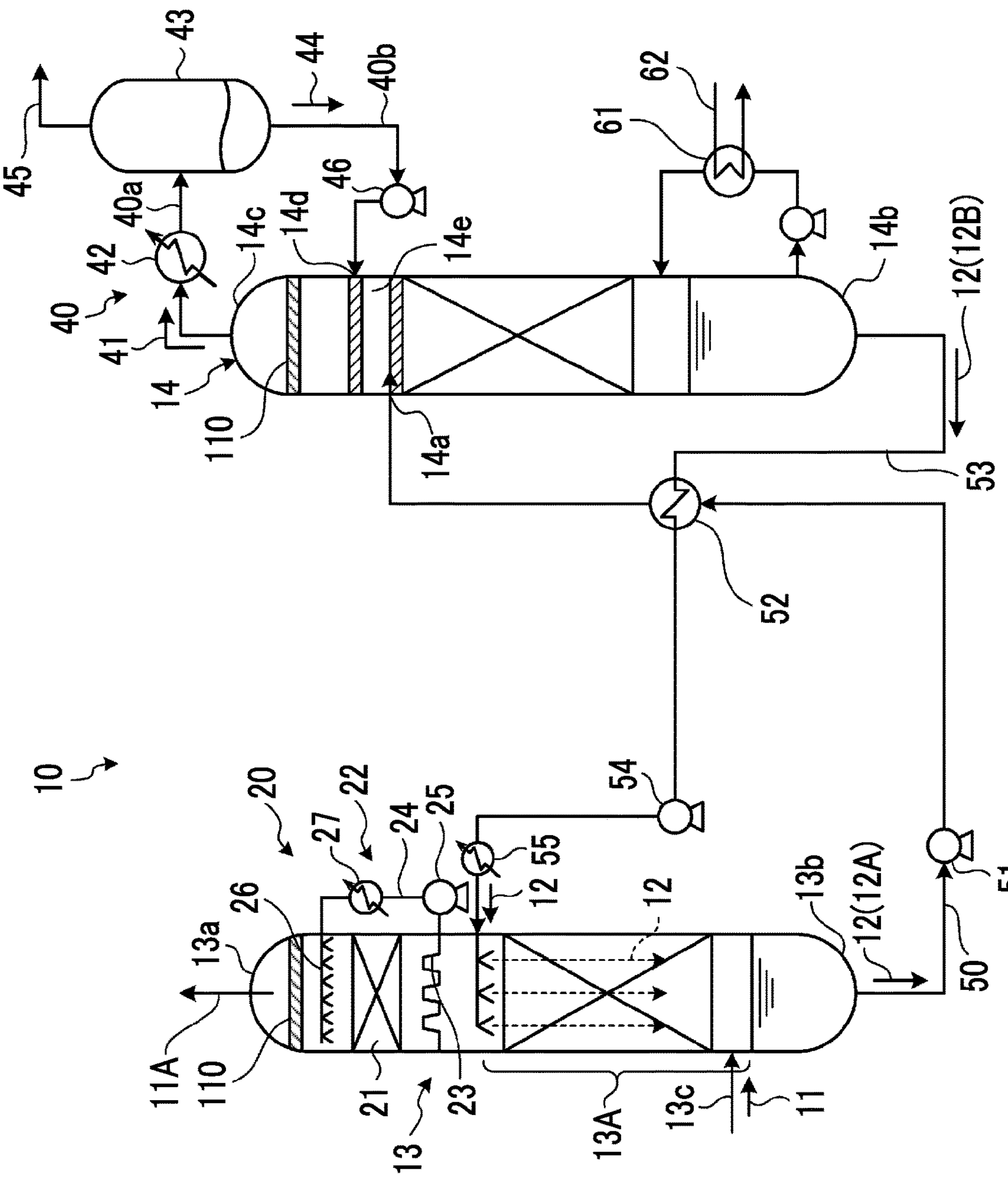
FIG. 1


10
11
11A
12
12(12A)
12(12B)
13
13A
13a
13b
13c
14
14a
14b
14c
14d
14e
20
21
22
23
24
25
26
27
40
40a
40b
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42
43
44
45
46
50
51
52
53
54
55
61
62
110

FIG. 2

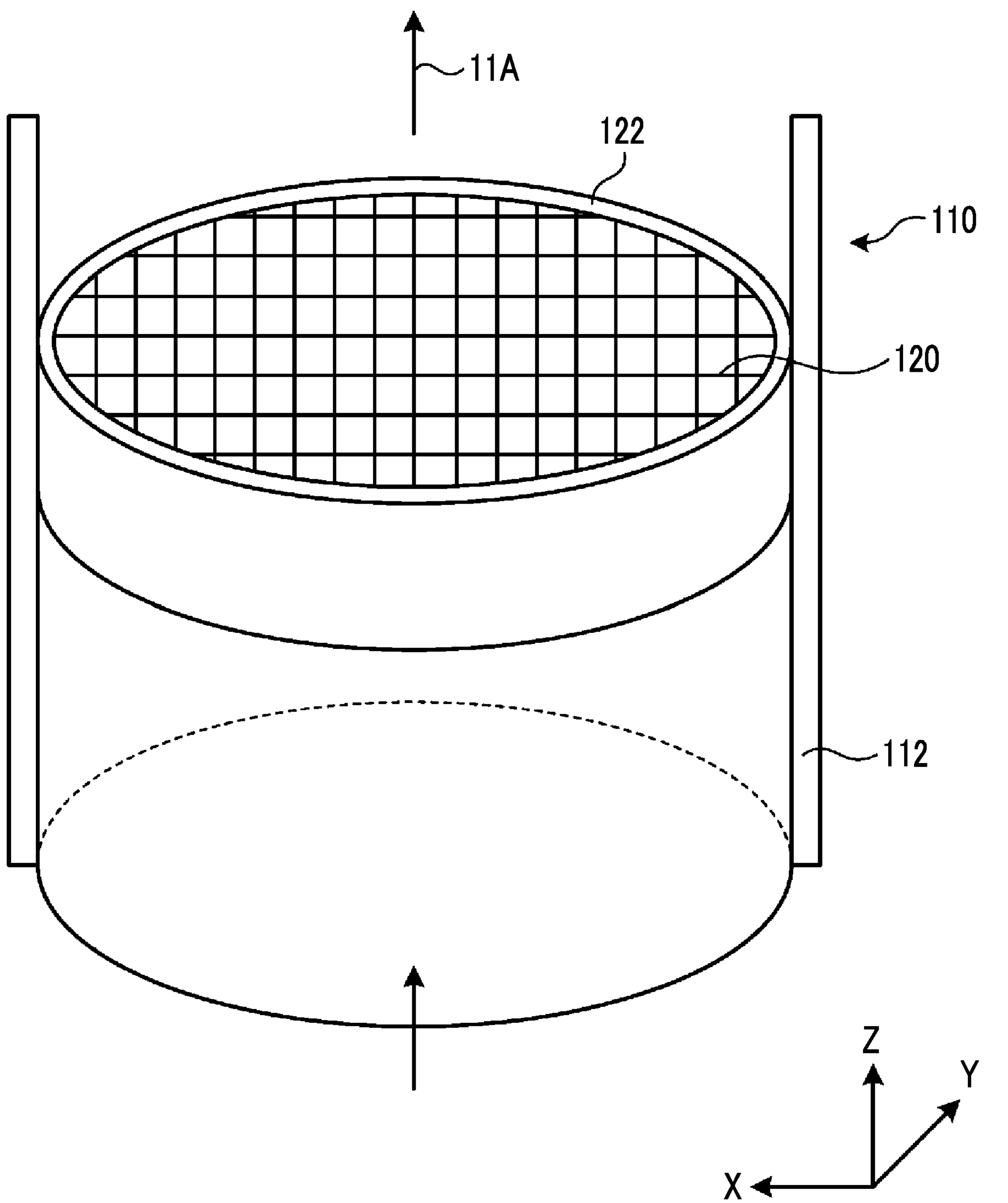


FIG. 3

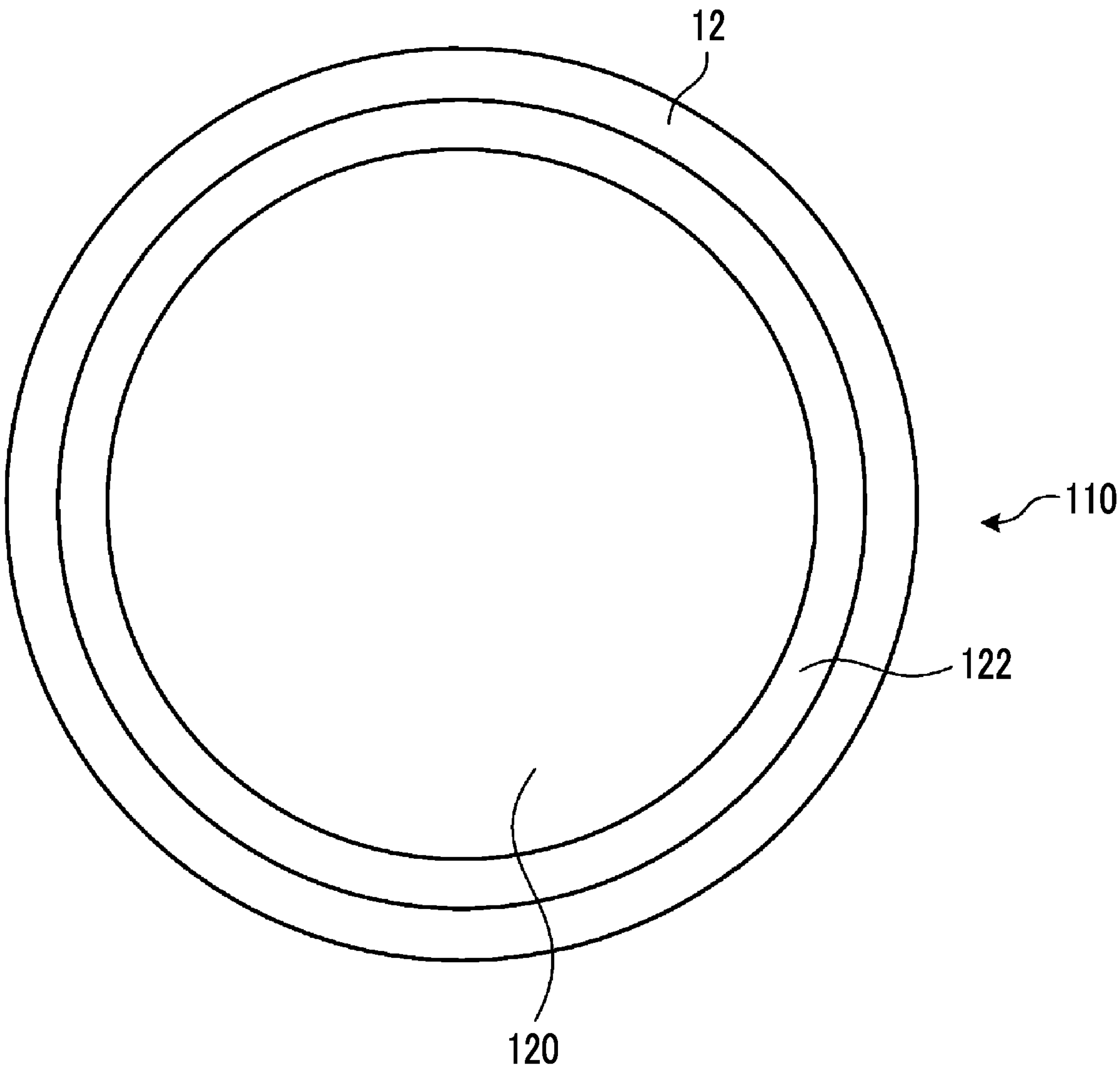


FIG. 4

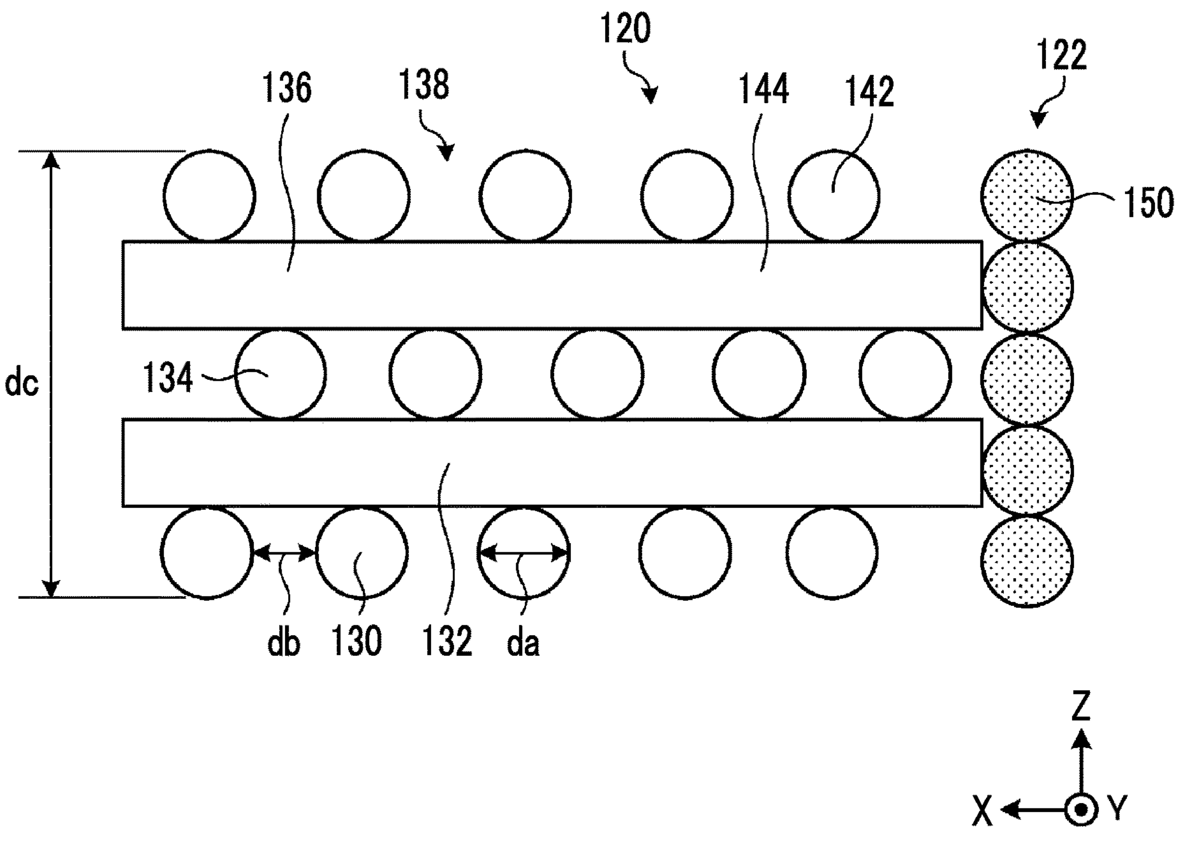


FIG. 5

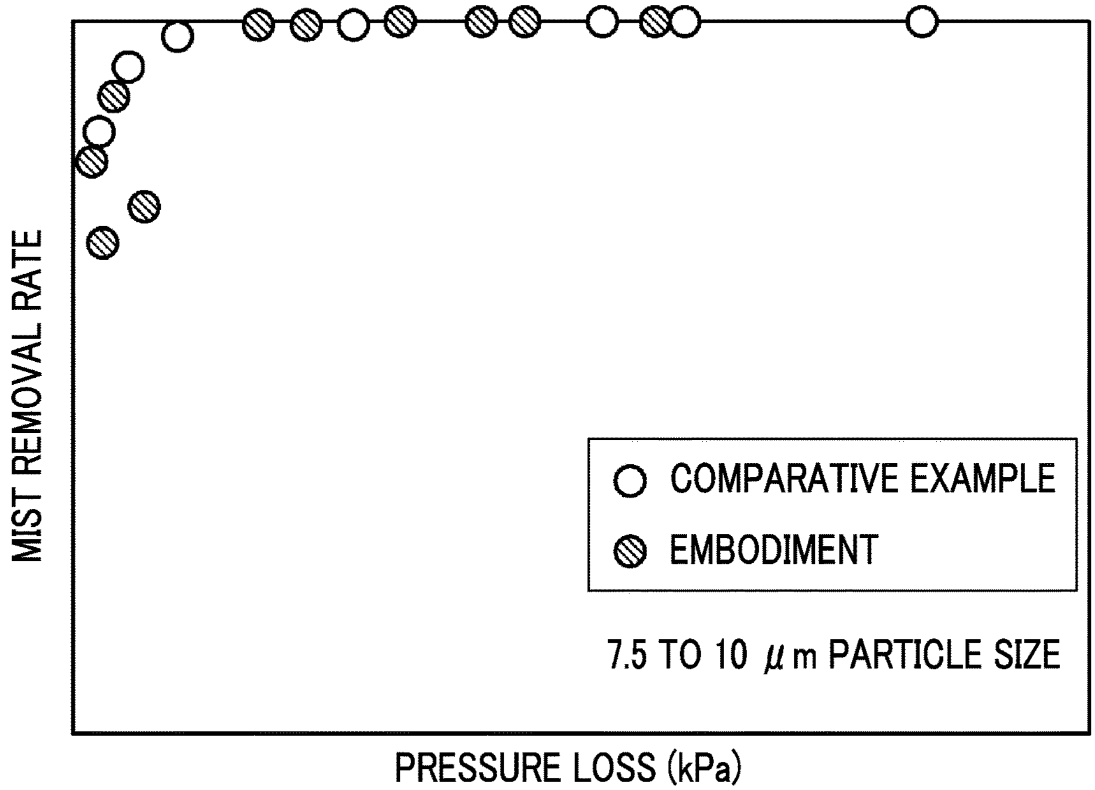

MIST REMOVAL RATE
COMPARATIVE EXAMPLE
EMBODIMENT
7.5 TO 10 μm PARTICLE SIZE
PRESSURE LOSS (kPa)

FIG. 6

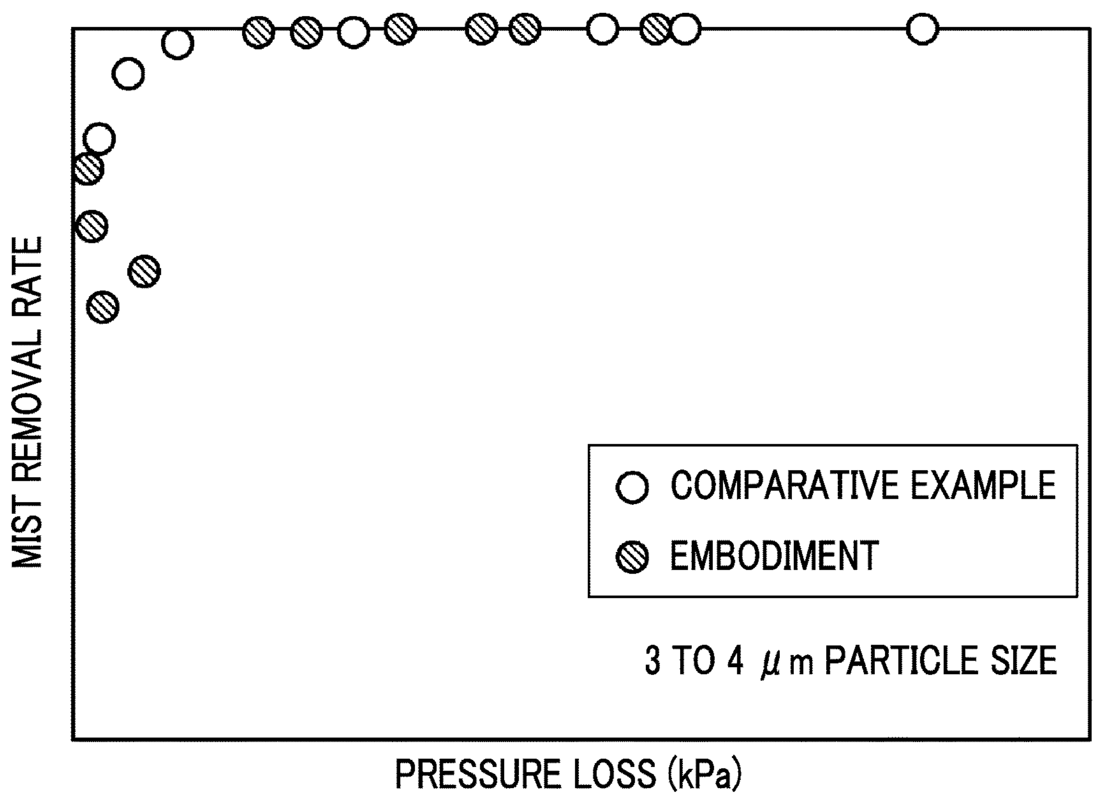

MIST REMOVAL RATE
COMPARATIVE EXAMPLE
EMBODIMENT
3 TO 4 μm PARTICLE SIZE
PRESSURE LOSS (kPa)

FIG. 7

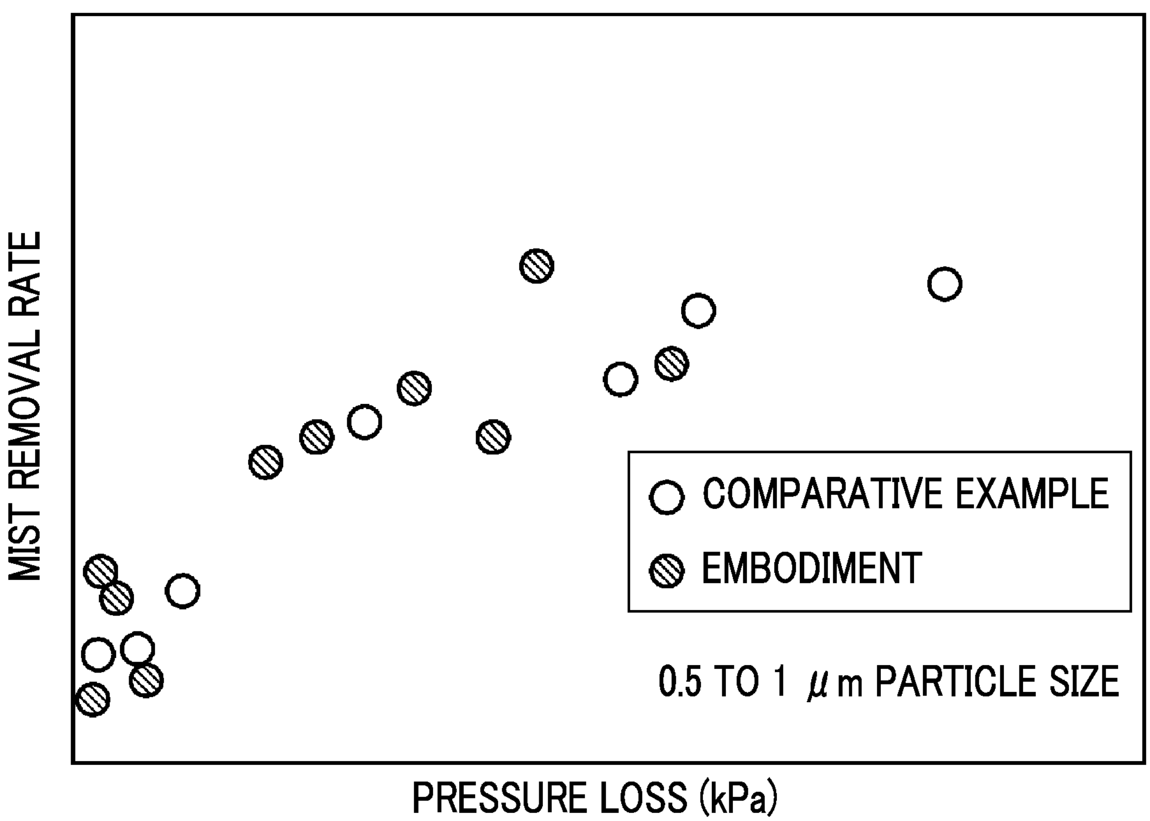
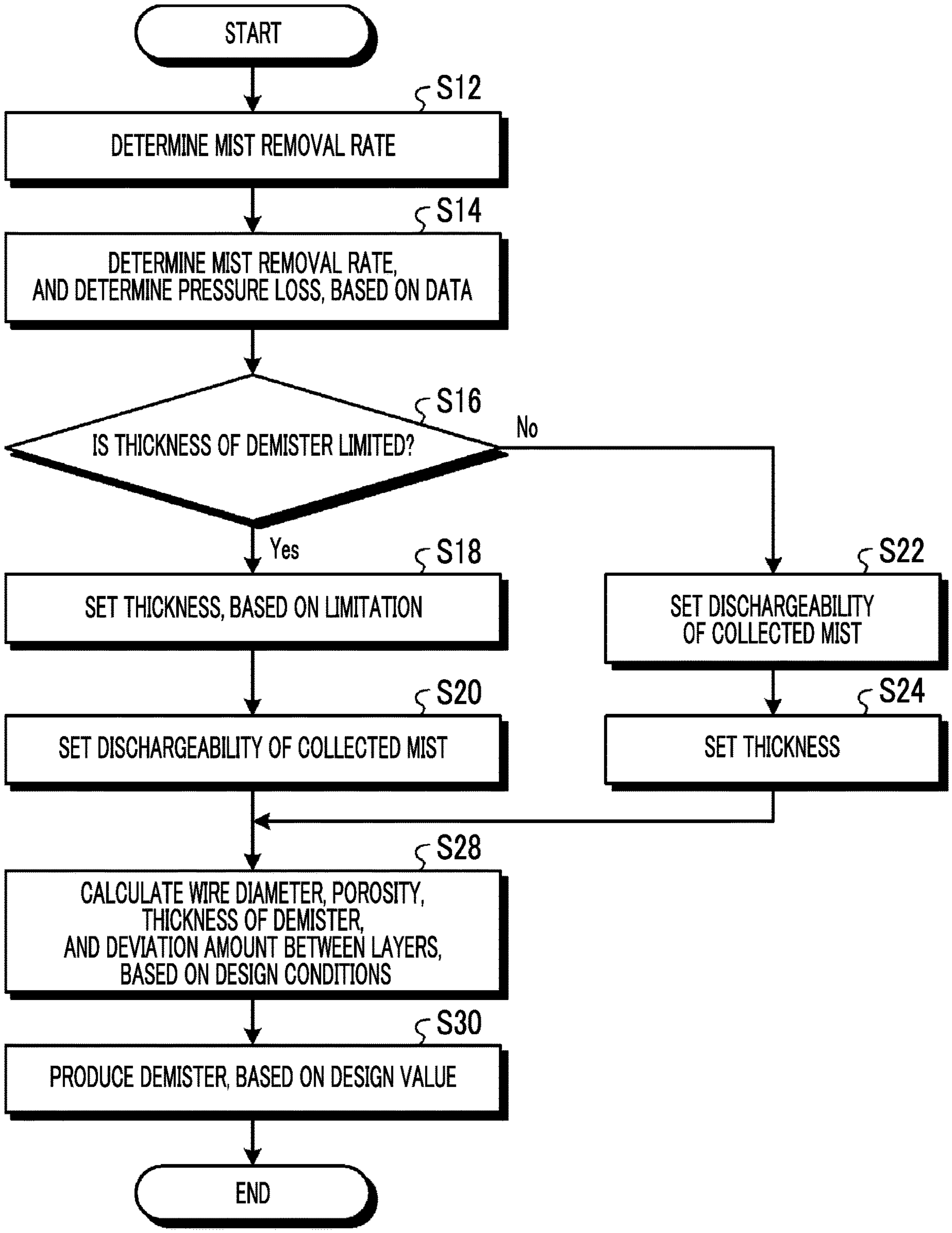

MIST REMOVAL RATE
COMPARATIVE EXAMPLE
EMBODIMENT
0.5 TO 1 μm PARTICLE SIZE
PRESSURE LOSS (kPa)

FIG. 8


START
S12
DETERMINE MIST REMOVAL RATE
S14
DETERMINE MIST REMOVAL RATE,
AND DETERMINE PRESSURE LOSS, BASED ON DATA
S16
IS THICKNESS OF DEMISTER LIMITED?
No
Yes
S18
SET THICKNESS, BASED ON LIMITATION
S20
SET DISCHARGEABILITY OF COLLECTED MIST
S22
SET DISCHARGEABILITY
OF COLLECTED MIST
S24
SET THICKNESS
S28
CALCULATE WIRE DIAMETER, POROSITY,
THICKNESS OF DEMISTER,
AND DEVIATION AMOUNT BETWEEN LAYERS,
BASED ON DESIGN CONDITIONS
S30
PRODUCE DEMISTER, BASED ON DESIGN VALUE
END

DEMISTER, ABSORPTION LIQUID ABSORBING TOWER, AND DEMISTER PRODUCTION METHOD

TECHNICAL FIELD

The present invention relates to a demister for removing mist in a gas, an absorption liquid absorbing tower, and a demister production method.

BACKGROUND ART

In recent years, a greenhouse effect resulting from CO_2 has been pointed out as one of causes of a global warming phenomenon, and there is an international and urgent need for countermeasures against the greenhouse effect to protect a global environment. Sources of CO_2 include all human activity fields in which fossil fuels are combusted, and a demand for discharge suppressing of CO_2 tends to increase more than ever. As a result, a power generation facility such as a thermal power plant using a large amount of the fossil fuels has been targeted for the following research. In the research, a combustion exhaust gas of an industrial facility such as a boiler and a gas turbine is brought into contact with an amine-based CO_2 absorption liquid. In this manner, a method for removing and recovering CO_2 contained in the combustion exhaust gas and a flue gas treatment system that store the recovered CO_2 without releasing the recovered CO_2 to an atmosphere have been energetically researched.

Various CO_2 recovery devices performing steps of removing and recovering CO_2 from the combustion exhaust gas by using a CO_2 absorption liquid have been proposed. The steps of removing and recovering CO_2 include a step of bringing the combustion exhaust gas and the CO_2 absorption liquid into contact with each other in a CO_2 absorbing tower (hereinafter, simply referred to as an "absorbing tower"), and a step of reusing the CO_2 absorption liquid by heating the CO_2 absorption liquid in an absorption liquid regeneration tower (hereinafter, simply referred to as a "regeneration tower") that regenerates the CO_2 absorption liquid which absorbs CO_2, diffusing CO_2, regenerating the CO_2 absorption liquid, and circulating the CO_2 absorption liquid to the CO_2 absorbing tower.

In the absorbing tower, for example, the CO_2 absorption liquid containing an absorbent such as alkanolamine is used for countercurrent contact, and CO_2 contained in an exhaust gas is absorbed by the CO_2 absorption liquid by a chemical reaction (exothermic reaction) so that the exhaust gas from which CO_2 is removed is released outward of the system. The CO_2 absorption liquid which absorbs CO_2 is also called a rich solution. A pressure of the rich solution is raised by a pump. The rich solution is heated in a heat exchanger by a high-temperature CO_2 absorption liquid (lean solution) in which CO_2 is diffused and regenerated in the regeneration tower, and is supplied to the regeneration tower.

In the CO_2 recovery device, a demister that collects mist is disposed in an outlet of the absorbing tower to suppress a possibility that the CO_2 absorption liquid may be discharged outward of the system (for example, refer to PTL 1).

CITATION LIST

Patent Literature

[PTL 1] Japanese Unexamined Patent Application Publication No. 2014-500

SUMMARY OF INVENTION

Technical Problem

Here, a demister is disposed in an outlet of an absorbing tower through which a gas containing mist flows or an outlet of a regeneration tower. Since the demister is disposed in a flow path of the gas to collect the mist, the demister is resistance to a gas flow. Therefore, a pressure loss occurs. When the pressure loss of the demister is large, overall efficiency of a device is degraded. In addition, when the mist stays in the demister, the staying mist causes an increase in the pressure loss.

In view of the above-described problems, an object of the present invention is to provide a demister, an absorption liquid absorbing tower, and a demister production method which can efficiently collect mist.

Solution to Problem

According to the present disclosure, in order to solve the above-described problems, there is provided a demister for collecting mist containing a CO_2 absorption liquid. A plurality of laminates including a first layer in which a plurality of linear structures having a first direction serving as an axial direction are disposed in parallel in a second direction orthogonal to the first direction, and a second layer in which a plurality of linear structures having a direction serving as an axial direction and different from the first direction are disposed in parallel in a direction orthogonal to the axial direction, are laminated in a direction orthogonal to the first direction and the second direction.

According to the present disclosure, in order to solve the above-described problems, there is provided an absorption liquid absorbing tower including an absorbing tower body to which a gas containing CO_2 is supplied, an absorption liquid supply unit that supplies an absorption liquid to the absorbing tower body, and the demister which is disposed on a downstream side in a flow direction of the gas from an absorption liquid supply position of the absorption liquid supply unit of the absorbing tower body, and collects mist containing an absorption liquid containing CO_2.

According to the present disclosure, in order to solve the above-described problems, there is provided a demister production method including a step of setting a mist removal rate, a pressure loss, a thickness, and dischargeability of collected mist, a step of determining the number of laminated layers of structures of a grid structure, a wire diameter, a wire interval, and a deviation amount between respective layers, based on the mist removal rate, the pressure loss, the thickness, and the dischargeability of the collected mist which are set, and a step of laminating the structures to produce a demister, based on the number of the laminated layers, the wire diameter, the wire interval, and the deviation amount between the respective layers which are set.

Advantageous Effects of Invention

According to the present invention, the mist can be efficiently collected.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a schematic view of a CO_2 recovery device according to the present embodiment.

FIG. **2** is a perspective view illustrating a schematic configuration of a demister.

FIG. 3 is a top view of the demister illustrated in FIG. 2.

FIG. 4 is a sectional view of the demister illustrated in FIG. 2.

FIG. 5 is a view for describing an example of characteristics of the demister.

FIG. 6 is a view for describing an example of characteristics of the demister.

FIG. 7 is a view for describing an example of characteristics of the demister.

FIG. 8 is a flowchart illustrating an example of a demister production method.

DESCRIPTION OF EMBODIMENTS

Hereinafter, a preferred embodiment of the present invention will be described in detail with reference to the accompanying drawings. The present invention is not limited by the embodiment. In a case where there are a plurality of the embodiments, the present invention also includes a configuration adopted by combining the respective embodiments with each other.

FIG. **1** is a schematic view of a CO_2 recovery device according to the present embodiment. The CO_2 recovery device of the present embodiment uses a CO_2 absorbent as an absorbent that absorbs carbon dioxide (CO_2), removes CO_2 from a gas in a CO_2 absorbing tower, and regenerates a CO_2 absorption liquid in an absorption liquid regeneration tower.

As illustrated in FIG. **1**, in the CO_2 recovery device **10** according to the present embodiment includes a CO_2 absorbing tower (hereinafter, referred to as an "absorbing tower") **13** including a CO_2 absorption unit (hereinafter, referred to as an "absorption unit") **13**A into which an introduction gas (hereinafter, referred to as "gas") **11** containing CO_2 is introduced, and removing CO_2 by bringing CO_2 contained in the gas and a CO_2 absorption liquid **12** into contact with each other, an absorption liquid regeneration tower (hereinafter, referred to as a "regeneration tower") **14** that regenerates a rich solution **12**A which is the CO_2 absorption liquid **12** absorbing CO_2 as the CO_2 absorption liquid by using steam of a reboiler **61**, a rich solution supply line **50** that extracts the rich solution **12**A from the absorbing tower **13** and introduces the rich solution **12**A to the regeneration tower **14** side, and a lean solution supply line **53** that extracts a lean solution **12**B which is the CO_2 absorption liquid in which CO_2 regenerated in the regeneration tower **14** is diffused from the regeneration tower **14**, introduces the lean solution **12**B into the absorbing tower **13**, and reuses the lean solution **12**B as the CO_2 absorption liquid. As the CO_2 absorption liquid **12**, the rich solution **12**A absorbing CO_2 and the lean solution **12**B from which CO_2 is diffused are circulated and reused inside the CO_2 recovery device. In describing the CO_2 recovery device of the present embodiment, an outline thereof will be merely described, and attached equipment is partially omitted in the description.

In the absorbing tower **13**, the gas **11** containing CO_2 is supplied after being cooled by cooling water in a cooling part. The absorbing tower **13** brings the gas **11** introduced by a gas introduction line **13***c* into countercurrent contact with the CO_2 absorption liquid **12** containing an amine-based CO_2 absorption component, and the CO_2 contained in the gas **11** is absorbed into the CO_2 absorption liquid **12** by a chemical reaction. The absorbing tower **13** causes a demister **110** to collect mist of a CO_2 removed exhaust gas **11**A after CO_2 is removed, and thereafter, releases the CO_2 removed exhaust gas **11**A outward of the system from a top portion **13***a*. The demister **110** will be described later.

A washing unit **20** is provided between a CO_2 absorption unit **13**A and the demister **110** inside the absorbing tower **13**. The washing unit **20** includes a gas-liquid contact portion **21** and rinse water circulation means **22**. The gas-liquid contact portion **21** passes through the CO_2 absorption unit **13**A, and brings the gas **11** accompanied by the CO_2 absorption liquid **12** absorbing CO_2 into gas-liquid contact with rinse water, and collects the CO_2 absorption liquid **12** contained in the gas **11** as the rinse water. The washing liquid circulation means **22** includes a rinse water tray **23**, a circulation line **24**, a pump **25**, a rinse water supply unit **26**, and a cooling unit **27**. The rinse water tray **23** is disposed on an upstream side of a gas flow of the gas-liquid contact portion **21**, that is, on a lower side of the gas-liquid contact portion **21** in a vertical direction. The rinse water tray **23** passes through the gas-liquid contact portion **21**, and collects the fallen rinse water. The circulation line **24** connects the rinse water tray **23** and the rinse water supply unit **26** outside the absorbing tower **13**. The pump **25** is installed in the circulation line **24**, and transports the rinse water in a predetermined direction. The rinse water supply unit **26** and the gas-liquid contact portion **21** are disposed on the downstream side of the gas flow, that is, on an upper side of the gas-liquid contact portion **21** in the vertical direction. The rinse water supply unit **26** supplies a washing liquid supplied by the circulation line **24** into the absorbing tower **13**. The supplied washing liquid falls on the gas-liquid contact portion **21**. For example, the rinse water supply unit **26** injects and supplies the washing liquid onto a spray. The cooling unit **27** is installed in the circulation line **24**, and cools the rinse water. As described above, the washing unit **20** collects the absorption liquid contained in the gas **11** by supplying the washing liquid from above the gas-liquid contact portion, collecting the washing liquid from below, and circulating the washing liquid. In the gas passing through the washing unit **20**, a portion of the washing liquid including the absorption liquid is accompanied by the gas **11**. That is, the gas passing through the washing unit **20** is accompanied by the mist containing the CO_2 absorption liquid. The demister **110** collects the mist containing the CO_2 absorption liquid.

In addition, the CO_2 recovery device **10** extracts the rich solution **12**A absorbing CO_2 from a bottom portion **13***b* of the absorbing tower **13** by the rich solution supply line **50**. The CO_2 recovery device **10** raises the pressure of the rich solution **12**A by the rich solution pump **51**, and supplies the rich solution **12**A to the regeneration tower **14** after heating the rich solution **12**A with the lean solution **12**B regenerated in the regeneration tower **14**, in a rich-lean solution heat exchanger **52** provided in an intersection portion between the rich solution supply line **50** and the lean solution supply line **53**.

In the regeneration tower **14**, the rich solution **12**A is discharged into the regeneration tower **14** from a rich solution introduction portion **14***a* in the vicinity of an upper part. The regeneration tower **14** releases most of CO_2 inside the regeneration tower **14** by an endothermic reaction between the rich solution **12**A and steam supplied from a bottom portion and generated by the reboiler **61**. The CO_2 absorption liquid releasing some or most of CO_2 inside the regeneration tower **14** becomes a semi-lean solution. When the semi-lean solution reaches a bottom portion **14***b* of the regeneration tower **14**, the semi-lean solution becomes the CO_2 absorption liquid (lean solution) **12**B from which all of CO_2 are substantially removed. In the rich solution **12**A, a portion of the lean solution **12**B is heated by the reboiler **61** to which saturated steam **62** is supplied, and the steam is supplied into the regeneration tower **14**.

In addition, in the regeneration tower **14**, a carrier gas (hereinafter, referred to as a "carrier gas") **41** whose main components are steam and CO_2 released from the rich solution **12**A and the semi-lean solution inside the tower is discharged from a tower top portion **14***c* after the mist is collected by the demister **112**.

The carrier gas **41** is supplied to a regeneration tower condensing unit **40**. The regeneration tower condensing unit **40** condenses the steam by cooling the carrier gas with the cooler **42**, and separates the steam into regeneration tower condensed water (hereinafter, referred to as "condensed water") **44** and a CO_2 gas **45** with a gas-liquid separator **43**. The regeneration tower condensing unit **40** presses the separated CO_2 gas **45** into an oil field by using an enhanced oil recovery (EOR), for example, or stores the separated CO_2 gas **45** in an aquifer.

In addition, the regenerated CO_2 absorption liquid (lean solution) **12**B is extracted from the bottom portion **14***b* of the regeneration tower **14** by the lean solution supply line **53**, and is cooled by the rich solution **12**A in the rich-lean solution heat exchanger **52**. Subsequently, the pressure is raised by the lean solution pump **54**, and is further cooled by a lean solution cooler **55**. Thereafter, the CO_2 absorption liquid is supplied into the absorbing tower **13**.

In the present embodiment, the regeneration tower condensing unit **40** for condensing moisture from the carrier gas **41** discharged from the tower top portion **14***c* of the regeneration tower **14** is provided outside the regeneration tower. The regeneration tower condensing unit **40** includes a discharge line **40***a* that discharges the carrier gas **41** from the tower top portion **14***c* of the regeneration tower **14**, a cooler **42** interposed in the discharge line **40***a*, the gas-liquid separator **43** that separates the condensed water **44** in which the steam is condensed by the cooler **42** and the CO_2 gas **45**, a return line **40***b* that causes the condensed water **44** to return to a head portion side of the regeneration tower, and a return water circulation pump **46** interposed in the return line **40***b*. The condensed water **44** separated and returned from the carrier gas **41** by the gas-liquid separator **43** is introduced by the return water circulation pump **46** from a condensed water introduction portion **14***d* on the tower top portion **14***c* side from the rich solution introduction portion **14***a* of the regeneration tower **14**.

The CO_2 recovery device **10** introduces the gas **11** containing CO_2 into the absorbing tower **13**, and brings the CO_2 contained in the gas **11** and the CO_2 absorption liquid **12** into contact with each other to remove CO_2. The gas supplied to the absorbing tower **13** and brought into contact with the CO_2 absorption liquid **12** passes through the demister **110**, and is discharged outward of the system. In addition, the CO_2 recovery device **10** introduces the rich solution **12**A absorbing CO_2 into the regeneration tower **14**, and regenerates CO_2 by using the steam of the reboiler. The carrier gas **41** of the regeneration tower **14** passes through the demister **110**, and is supplied to the regeneration tower condensing unit **40**. The CO_2 recovery device **10** circulates and reuses the CO_2 absorption liquid **12** by using a circulation line between the absorbing tower **13** and the regeneration tower **14**. The CO_2 recovery device **10** condenses moisture from the carrier gas **41** accompanied by the separated CO_2 in the regeneration tower condensing unit **40**. The CO_2 recovery device **10** cools the carrier gas **41** to separate the condensed water **44** in which the steam is condensed and the CO_2 gas **45**. The CO_2 recovery device **10** causes the condensed water **44** to return to the tower top portion **14***c* side of the rich solution introduction portion **14***a* into which the rich solution **12**A of the regeneration tower **14** is introduced, and supplies the condensed water **44**.

FIG. **2** is a perspective view illustrating a schematic configuration of the demister. FIG. **3** is a top view of the demister illustrated in FIG. **2**. FIG. **4** is a sectional view of the demister illustrated in FIG. **2**. Hereinafter, a case where the demister **110** is installed in the absorbing tower will be described. The demister **110** is disposed in a circular absorbing tower body **112**. The absorbing tower body **112** is a flow path of the absorbing tower through which the exhaust gas **11**A flows. The absorbing tower body **112** of the present embodiment has a cylindrical shape having a circular cross section. A shape of the absorbing tower body **112** is not limited thereto, and a cross section may be rectangular, for example.

The demister **110** includes a laminate unit **120** and a holding unit **122**. The laminate unit **120** has a first layer **130**, a second layer **132**, a third layer **134**, a fourth layer **136**, and a fifth layer **138**. In the laminate unit **120**, the first layer **130**, the second layer **132**, the third layer **134**, the fourth layer **136**, and the fifth layer **138** are laminated in a traveling direction of the exhaust gas **11**A. That is, each layer extends in a plane orthogonal to the traveling direction of the exhaust gas **11**A. Each layer of the laminate unit **120** is disposed on an entire inner surface of the absorbing tower body **112**, and the exhaust gas **11**A passes through each layer when passing through the demister **110**. In the present embodiment, the exhaust gas **11**A passes through the first layer **130**, the second layer **132**, the third layer **134**, the fourth layer **136**, and the fifth layer **138** in this order. In the demister **110** of the present embodiment, the first layer **130** and the second layer **132** are one laminate, and the third layer **134** and the fourth layer **136** are one laminate. In the present embodiment, a structure is formed so that five layers are laminated. However, without being limited, the number of laminated layers may be six or more. In the demister **110**, the number of layers to be laminated may be an odd number or an even number as in the present embodiment. In the laminate unit **120**, the laminates formed by two layers have the same configuration, and a plurality of the laminates are laminated. That is, in the laminate unit **120**, the odd-numbered layers have the same configuration, and the even-numbered layers have the same configuration. In the present embodiment, two layers having different structures are alternately laminated. However, even when three types of layers are provided and three types of layers are laminated in order, four or more types of layers may be provided and four types of layers may be laminated in order.

The first layer **130** has a plurality of linear structures **142**. The structure **142** is a straight rod-shaped member. The structure **142** is formed of a material that is not corroded by the exhaust gas **11**A, for example, a resin such as polypropylene or metal such as stainless steel. In the first layer **130**, the plurality of structures **142** whose first direction is the axial direction are disposed in parallel in a second direction orthogonal to the first direction. The first direction and the second direction are in-plane directions orthogonal to the traveling direction of the exhaust gas **11**A. In the plurality of structures **142** of the present embodiment, a wire diameter da and an interval db between the structures adjacent to each other are constant. That is, in the first layer **130**, the structures **142** having the same diameter are disposed at a constant interval.

The second layer **132** has a plurality of linear structures **144**. The structure **144** is a straight rod-shaped member similar to the structure **142**. The structure **144** is formed of a material that is not corroded by the exhaust gas **11**A, for example, a resin such as polypropylene or metal such as stainless steel. In the second layer **132**, the plurality of structures **144** whose second direction is the axial direction are disposed in parallel in the first direction. In the plurality of structures **144**, a wire diameter and an interval between the structures adjacent to each other are constant. That is, in the second layer **132**, the structures **144** having the same diameter da are disposed at a constant interval db.

In the third layer **134** and the fifth layer **138**, as in the first layer **130**, a plurality of structures **142** whose first direction is the axial direction are disposed in parallel in the second direction orthogonal to the first direction. In the fourth layer **136**, as in the second layer **132**, the plurality of structures **142** whose second direction is the axial direction are disposed in parallel in the first direction.

The laminate unit **120** is disposed so that a direction in which the structure **142** and the structure **144** of two adjacent layers are orthogonal to each other is the axial direction. That is, the laminate unit **120** is a grid structure in which the laminate configured to include two layers has the structures **142** and **144**. In addition, in the laminate unit **120** of the present embodiment, disposition positions of the structures **142** of the first layer **130**, the third layer **134**, and the fifth layer **138** are the same positions on a plane orthogonal to the traveling direction of the exhaust gas **11**A. In the laminate unit **120** of the present embodiment, disposition positions of the structures **144** of the second layer **132** and the fourth layer **136** are the same positions on the plane orthogonal to the traveling direction of the exhaust gas **11**A. The length of the laminate unit **120** in the traveling direction of the exhaust gas **11**A is a thickness dc.

The holding unit **122** is disposed on the entire outer periphery of the laminate unit **120**. The holding unit **122** is fixed to the absorbing tower body **112**, and supports the laminate unit **120**. The holding unit **122** may be supported by a rib provided on an inner wall of the absorbing tower body **112**, or may be supported by means of screwing.

In the holding unit **122**, the structure **150** is disposed corresponding to each layer. The structure **150** is formed of a material that is not corroded by the exhaust gas **11**A, for example, a resin such as polypropylene or metal such as stainless steel. It is preferable that the structure **150** is formed of a material the same as that of the structures **142** and **144** and is integrally produced. The structure **150** is disposed in all layers of the laminate unit **120**. The structure **150** is fixed to the structure **150** of the adjacent layer. As a fixing method, the structures **150** may be melted and fixed, or may be fixed to each other by an adhesive agent or by means of welding. The structure **150** is connected to end portions of the plurality of structures **142** and **144** in the disposed layers, and holds the plurality of structures **142** and **144** in each layer in predetermined positions.

Here, the laminate unit **120** of the present embodiment adopts a structure in which the disposition positions of the structures **142** of the first layer **130**, the third layer **134**, and the fifth layer **138** are the same positions, and the disposition positions of the structures **144** of the second layer **132** and the fourth layer **136** are the same positions on the plane orthogonal to the traveling direction of the exhaust gas **11**A. However, the present invention is not limited thereto. In the laminate unit **120**, the disposition positions of the structures **142** of the first layer **130**, the third layer **134**, and the fifth layer **138** on the plane orthogonal to the traveling direction of the exhaust gas **11**A may be deviated between the layers. In addition, in the laminate unit **120**, the disposition positions of the structures **144** of the second layer **132** and the fourth layer **136** on the plane orthogonal to the traveling direction of the exhaust gas **11**A may be deviated between the layers.

The demister **110** can efficiently collect the mist by deviating the positions of the structures between the layers of the laminate unit **120**. In the demister **110**, the pressure loss increases by deviating the positions of the structures between the layers of the laminate unit **120**.

The demister **110** is provided with the laminate unit **120**, and the structure **142** and **144** of the laminate in the two adjacent layers have a grid structure. In this manner, performance of the demister can be easily controlled. In this manner, the pressure loss can be highly accurately controlled.

FIGS. **5** to **7** are views for describing an example of characteristics of the demister. FIG. **5** is a graph illustrating a result obtained by evaluating a relationship between the pressure loss and the mist removal rate when a grain size of the mist contained in the carrier gas is set to 7.5 μm or larger and 10 μm or smaller. FIG. **6** is a graph illustrating a result obtained by evaluating a relationship between the pressure loss and the mist removal rate when the grain size of the mist contained in the carrier gas is set to 3 μm or larger and 4 μm or smaller. FIG. **7** is a graph illustrating a result obtained by evaluating a relationship between the pressure loss and the mist removal rate when the grain size of the mist contained in the carrier gas is set to 0.5 μm or larger and 1.0 μm or smaller. In each of FIGS. **5** to **7**, a horizontal axis represents the pressure loss (kPa), and a vertical axis represents the mist removal rate (%). Here, the mist removal rate is the amount of the mist (moisture) removed by passing through the demister **110**. The mist removal rate (%) is calculated by ((number of entering water particles-number of water particles after passing through the demister)/(number of entering water particles))×100. The graphs illustrated in FIGS. **5** to **7** illustrate the evaluation results in which the mist removal rate is evaluated by changing the number of laminated layers and the thickness of the structure and setting the pressure loss as various values in the demister of the grid structure of the present embodiment. In addition, as a comparative example, the mist removal rate is evaluated by setting the pressure loss as various values even in a demister having a structure in the related art. The demister in the comparative example has a structure in which the structures for collecting the mist are not disposed in an aligned manner.

As illustrated in FIGS. **5** to **7**, it can be understood that the demister **110** has a correlation between the pressure loss and the mist removal rate. It can be understood that the correlation is similar to that of the structure in the related art. The demister **110** of the present embodiment is a demister including required performance of the mist removal rate and the pressure loss by designing the laminate unit **120**, based on a target mist removal rate and a target pressure loss. In addition, the demister **110** can highly accurately calculate the pressure loss of the demister by forming a grid structure in which layers having a constant wire diameter and interval of the structure are superimposed. In this manner, the demister **110** that realizes the target mist removal rate at the time of design can be highly accurately designed and produced. Accuracy in mist removal performance of the demister **110** can be improved. Accordingly, a margin to be added due to production errors can be reduced. In this manner, the CO_2 recovery device **10** can realize the required mist removal rate while reducing the pressure loss in the demister **110**. The pressure loss of the demister **110** can be reduced. Accordingly, the thickness dc of the demister **110** can be reduced. In addition, the demister **110** can reduce the thickness dc of the demister **110** by increasing the pressure loss per unit thickness.

In addition, the demister **110** adopts a grid structure in which structures are disposed in parallel in each layer, and are orthogonal to each other between the adjacent layers. Accordingly, it is possible to suppress a possibility that the collected mist may be stored inside the laminate unit **120**. That is, it is possible to adopt a structure in which the demister **110** is less likely to be clogged. In this manner, it is possible to suppress a possibility that the pressure loss may fluctuate when in use, and a stable operation can be performed.

In addition, as described above, in the demister **110**, it is preferable that the positions of the structures disposed in the same direction of the adjacent laminates are deviated on the plane orthogonal to the flow direction of the gas. That is, in the demister **110**, it is preferable that the position of the structure of the laminate is deviated from at least one of other laminates in a direction in which the structures are aligned. In this manner, the pressure loss per thickness dc of the demister **110** can be increased, and the thickness of the demister **110** can be reduced.

In addition, the demister **110** can adjust porosity or space occupancy of the structure by adjusting the diameter da and the interval db of the structure. The demister **110** can reduce occurrence of clogging in the layer by increasing the porosity.

In the laminate unit **130** of the present embodiment, the wire diameter da of the plurality of structures **142** and **144** and the interval db between the structures adjacent to each other are respectively constant. However, the present invention is not limited thereto. The plurality of structures **142** and **144** may be respectively disposed in parallel, and may include structures having different wire diameters da, or may include structures having the intervals db with different distances. As in the present embodiment, the wire diameter and the interval are set to be constant. Accordingly, production can be facilitated, design can be facilitated, and performance can be stabilized. In addition, here, a fact that the wire diameter and the interval are constant means that both are constant in terms of a value which allows a difference generated due to production errors. In addition, as described above, the laminate is not limited to a case of the structure having the two layers, and may be a structure in which three or more layers are laminated. For example, in a case of the structure having three layers, the axial directions of the structures of the respective layers may be differently changed by 30°.

Next, a demister production method will be described. FIG. **8** is a flowchart illustrating an example of the demister production method. A process illustrated in FIG. **8** may be performed by a worker carrying out each work or detecting data input from a processing apparatus or a production apparatus.

In the demister production method, the mist removal rate is determined (Step S**12**). In the demister production method, the pressure loss is calculated, based on the mist removal rate and data (Step S**14**). In the demister production method, it is determined whether the thickness of the demister is limited (Step S**16**).

In the demister production method, when it is determined that the thickness of the demister is limited (Yes in Step S**16**), the thickness is set, based on the limitation (Step S**18**), and dischargeability of the collected mist is set (Step S**20**). The dischargeability of the collected mist means whether it is easy to discharge the mist collected by the laminate unit outward of the demister. In the demister production method, the thickness is set based on the limitation, and the dischargeability of the collected mist is set within a range that can be set by the thickness.

In the demister production method, when it is determined that the thickness of the demister is not limited (No in Step S**16**), the dischargeability of the collected mist is set (Step S**22**), and the thickness is set, based on the limitation (Step S**24**). In the demister production method, the dischargeability of the collected mist is set, and the thickness is set within a range that satisfies the dischargeability of the collected mist.

In the demister production method, after the thickness and the dischargeability of the collected mist are set, the wire diameter, the porosity, the thickness of the demister, and the deviation amount between the layers are calculated, based on design conditions (Step S**28**). That is, the number of laminated layers of the demister, the wire diameter (diameter) of the structure in each layer, the porosity, the thickness of the demister, and the deviation amount of the structures between the layers are calculated to satisfy the mist removal rate, the pressure loss, the thickness, and the dischargeability of collected mist which are to be set. The wire diameter (diameter) of the structure and the porosity are determined. Accordingly, the interval between the structures can be calculated.

In the demister production method, the demister is produced, based on a design value (Step S**30**). For example, the demister can be produced by a three-dimensional modeling apparatus. In addition, the demister can also be produced by disposing and fixing the structure at a designed position.

In the demister production method, the number of laminated layers of the laminate unit of the grid structure, the wire diameter, and the deviation amount are designed, and the demister is produced, based on the design. In this manner, the demister close to a designed numerical value can be highly accurately produced. In this manner, it is possible to obtain the demister whose pressure loss is close to the design value and whose mist removal rate satisfies a target. In addition, the demister is produced by the three-dimensional modeling apparatus. Accordingly, the structure having the constant interval can be automatically and highly accurately produced. In addition, from a viewpoint of facilitating the design and facilitating the production, it is preferable that the wire diameter and the interval in one layer are constant as in the above-described embodiment. However, the wire diameter and the interval may vary depending on the positions.

REFERENCE SIGNS LIST

10: CO_2 recovery device
11: Introduction gas (gas)
12 CO_2 absorption liquid
12A: Rich solution
12B: Lean solution
13A: CO_2 absorption unit
13: CO_2 absorbing tower
14: Absorption liquid regeneration tower
41: Carrier gas
42: Cooler
43: Gas-liquid separator
44: Regeneration tower condensed water
45: CO_2 gas
46: Return water circulation pump
50: Rich solution supply line
51: Rich solution pump

52: Rich-lean solution heat exchanger
53: Lean solution supply line
100, **110**: Demister
112: Absorbing tower body
120: Laminate unit
122: Holding unit
130: First layer
132: Second layer
134: Third layer
136: Fourth layer
138: Fifth layer
150, **142**, **144**: Structure
da: Diameter
db: Interval
dc: Thickness

The invention claimed is:

1. An absorption liquid absorbing tower comprising:

an absorbing tower body to which a gas containing CO_2 is supplied; and

a demister for collecting mist containing a CO_2 absorption liquid,

wherein the demister

disposed between an inlet into which the gas is supplied and an outlet into which the CO_2-removed flue gas obtained by removing the CO_2 from the gas is discharged in the absorption tower body,

is a plurality of laminates including a first layer in which a plurality of linear structures having a first direction serving as an axial direction are disposed in parallel in a second direction orthogonal to the first direction, and a second layer in which a plurality of linear structures having a direction serving as an axial direction and different from the first direction are disposed in parallel in a direction orthogonal to the axial direction, are laminated in a direction orthogonal to the first direction and the second direction,

in the first layer, a position of the structure in the second direction is deviated from the structure of the first layer of at least one of other laminates, and

the demister has a grid structure in which layers having a constant wire diameter and interval of the structure are superimposed,

a laminating direction of the plurality of laminates is a direction from the inlet toward the outlet.

2. An absorption liquid absorbing tower according to claim **1**,

wherein in the first layer, a wire diameter of the structure and an interval between the structures adjacent to each other are constant, and

in the second layer, a wire diameter of the structure and an interval between the structures adjacent to each other are constant.

3. An absorption liquid absorbing tower according to claim **1**,

wherein in the second layer, the second direction of the structure is the axial direction.

4. An absorption liquid absorbing tower according to claim **1**, further comprising:

a holding unit disposed around the laminate and supporting end portions of the structures of the first layer and the second layer.

5. An absorption liquid absorbing tower according to claim **1** further comprising:

an absorption liquid supply unit that supplies an absorption liquid to the absorbing tower body;

the demister is disposed on a downstream side in a flow direction of the gas from an absorption liquid supply position of the absorption liquid supply unit of the absorbing tower body.

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